

ATPL EASA STUDY GUIDE | 033 - FLIGHT PLANNING

Chapter 6: Standard Departures (SID), STARs & Holding Entries

Syllabus Source: CAE Oxford (pp. 209-248) | Pass Target: >= 90% AviationExam

1. STANDARD INSTRUMENT DEPARTURES (SID) & TRANSITIONS

A SID links the take-off runway to the en-route airway structure, ensuring safe obstacle clearance:

- Design Climb Gradient (PDG): Standard ICAO procedure design climb gradient is 3.3% (approx 200 ft/NM). If close-in obstacles exist, a steeper gradient (e.g. 5.5%) is published with an altitude up to which it must be maintained.
- Altitude Restrictions: At or Above (bar under number: 5000), At or Below (bar over number), Mandatory Crossing Altitude (bars above and below), Altitude Window (e.g. FL 110 - FL 130).

2. HOLDING PATTERN GEOMETRY & STANDARD SPEEDS

Standard holding pattern is an oval with RIGHT-HAND TURNS (non-standard is left-hand turns):

- Leg Timings: Inbound leg timing: 1.0 MINUTE at or below 14,000 ft MSL; 1.5 MINUTES above 14,000 ft (FL 140).
- ICAO Maximum Holding Speeds: Up to 14,000 ft: 230 kt (turboprops 170 kt); 14,000 to 20,000 ft: 240 kt; Above FL 200: 265 kt (or 0.83 Mach).

3. THE THREE HOLDING ENTRY SECTORS

Entry Sector	Angular Width & Relative Heading	Standard Entry Maneuver Procedure
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Sector 1: Parallel Entry	110° sector on the holding side.	Fly to fix, turn to parallel the outbound track (heading reciprocal of inbound) on the non-holding side for 1 min, turn left toward holding side to intercept inbound track.
Sector 2: Teardrop (Offset)	70° sector on the non-holding side.	Fly to fix, turn outbound at 30° OFFSET into the holding area for 1 min, then turn right to intercept the inbound track.
Sector 3: Direct Entry	180° sector (the entire remaining half).	Fly directly to the fix, turn right onto the outbound heading, and enter the race-track pattern directly!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): SID Climb Gradient & Rate of Climb

Formula: Rate of Climb (ft/min) = Groundspeed (kt) x Climb Gradient (%). Example: A SID specifies a 5.0% gradient to 6,000 ft. Your departure Groundspeed is 180 kt. - Step 1: ROC = $180 \times 5.0 = 900$ ft/min on the VSI. - Step 2: If tailwind increases groundspeed to 210 kt: ROC = $210 \times 5.0 = 1,050$ ft/min required! Rule: Higher groundspeed requires a higher vertical speed to maintain the same flight path gradient!

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Determining Holding Entry Sector**

SCENARIO (Classic ATPL Exam Question): An aircraft is flying on a heading of 350° toward VOR 'ZUR'. ATC instructs: 'Hold at ZUR on the 090° radial, inbound track 270°, right-hand turns.' QUESTION: What is the correct ICAO holding entry procedure? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify Inbound Track and Turn Direction: - Inbound Track = 270° - Turn direction = Standard RIGHT-HAND turns. - Outbound Track = 090° (reciprocal of 270°). Step 2: Draw the 3 Entry Sectors relative to Inbound Track (270°): - For standard right-hand turns: * Sector 1 (Parallel - 110°): From 270° to $(270° - 110°) = 270°$ to 160° clockwise (160° to 270°). * Sector 2 (Teardrop - 70°): From 270° to $(270° + 70°) = 270°$ to 340° (clockwise 270° to 340°). * Sector 3 (Direct - 180°): From 340° to 160° (the entire remaining arc: 340° through North/360° to 160°). Step 3: Check Aircraft Inbound Heading (350°): - Aircraft is arriving on heading 350°. - Compare 350° with the sectors: * 350° falls squarely inside Sector 3 (which spans from 340° through 360° to 160°)! FINAL ANSWER: Sector 3 (DIRECT ENTRY)! Upon reaching ZUR, the pilot turns right directly onto the outbound heading of 090°.

[!] EXAM TRAP (AVIATIONEXAM) - The Right-Hand vs Left-Hand Holding Rule

- For RIGHT-HAND turns: Sector 1 (Parallel) is on the left (110°); Sector 2 (Teardrop) is on the right (70°). - For LEFT-HAND turns: The sectors are MIRRORED! Sector 1 is 110° on the right, Sector 2 is 70° on the left! - Sector 3 (Direct) is always the large 180° sector opposite the entry.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the standard outbound leg TIMING for a holding pattern at or below 14,000 ft MSL?**

- [A]: 1.5 minutes
- [B]: 1.0 MINUTE (60 seconds)
- [C]: 2.0 minutes
- [D]: 45 seconds

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ICAO PANS-OPS, the outbound leg is timed for 1.0 minute up to and including 14,000 ft (FL 140), and 1.5 minutes above 14,000 ft.

Q2: In a standard right-hand holding pattern, what is the angular width of SECTOR 1 (Parallel Entry)?

- [A]: 70°
- [B]: 110 DEGREES

- [C]: 180°
- [D]: 90°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Sector 1 (Parallel) covers an arc of 110°; Sector 2 (Teardrop) covers 70°; Sector 3 (Direct) covers 180°.

Q3: What is the maximum ICAO holding speed for aircraft holding at 10,000 ft MSL?

- [A]: 250 kt
- [B]: 230 KNOTS (or 170 kt for normal turboprops)
- [C]: 200 kt
- [D]: 265 kt

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Doc 8168, maximum holding speed up to 14,000 ft is 230 kt for jet aircraft.